

Here is a clean, relational database structure for a universal e-commerce platform. This schema is designed to be scalable and normalizes the data to prevent redundancy, which is especially helpful when mapping these tables to an ORM (like Django's or SQLAlchemy) for full-stack development.

Core Entities & Relationships

1. Users

The central table for customer authentication and profile data.

Column Name	Data Type	Constraints / Relations
id	UUID / Integer	Primary Key, Auto-increment
email	String	Unique, Not Null
password_hash	String	Not Null
first_name	String	
last_name	String	
created_at	Timestamp	Default: Current Time

2. Categories

Allows products to be grouped. Including a parent_id allows for sub-categories (e.g., Electronics -> Laptops).

Column Name	Data Type	Constraints / Relations
id	Integer	Primary Key, Auto-increment
name	String	Not Null
slug	String	Unique (for URL routing)

Column Name	Data Type	Constraints / Relations
parent_id	Integer	Foreign Key (Categories.id), Nullable

3. Products

The core catalog.

Column Name	Data Type	Constraints / Relations
id	UUID / Integer	Primary Key, Auto-increment
category_id	Integer	Foreign Key (Categories.id)
name	String	Not Null
description	Text	
price	Decimal(10, 2)	Not Null
stock_quantity	Integer	Default: 0
is_active	Boolean	Default: True (to hide discontinued items)
created_at	Timestamp	Default: Current Time

4. Reviews (Ratings & Comments)

Combines both the star rating and the text comment into a single entity to ensure a user's comment is always tied to their rating.

Column Name	Data Type	Constraints / Relations
id	Integer	Primary Key, Auto-increment
product_id	UUID / Integer	Foreign Key (Products.id)

Column Name	Data Type	Constraints / Relations
user_id	UUID / Integer	Foreign Key (Users.id)
rating	Integer	Check: 1 to 5, Not Null
comment	Text	Nullable
created_at	Timestamp	Default: Current Time

(Note: You would typically add a Unique Constraint on [product_id, user_id] so a user can only leave one review per product).

5. Saved Items (Wishlist)

A junction table linking users to the products they want to save for later.

Column Name	Data Type	Constraints / Relations
id	Integer	Primary Key, Auto-increment
user_id	UUID / Integer	Foreign Key (Users.id)
product_id	UUID / Integer	Foreign Key (Products.id)
added_at	Timestamp	Default: Current Time

6. Orders

Tracks the overarching checkout event for a user.

Column Name	Data Type	Constraints / Relations
id	UUID / Integer	Primary Key, Auto-increment
user_id	UUID / Integer	Foreign Key (Users.id)

Column Name	Data Type	Constraints / Relations
status	Enum/String	'Pending', 'Shipped', 'Delivered', 'Cancelled'
total_amount	Decimal(10, 2)	Not Null
shipping_address	Text	Not Null
created_at	Timestamp	Default: Current Time

7. Order Items

A bridge table that records the specific products bought within a single order. It is crucial to store the price_at_purchase here, as the live product price might change in the future, but historical order records must remain accurate.

Column Name	Data Type	Constraints / Relations
id	Integer	Primary Key, Auto-increment
order_id	UUID / Integer	Foreign Key (Orders.id)
product_id	UUID / Integer	Foreign Key (Products.id)
quantity	Integer	Check: > 0
price_at_purchase	Decimal(10, 2)	Not Null